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Cite this: *Org. Chem. Front.*, 2020, **7**, 2975

Photoinduced double [2 + 2] cycloaddition relay of yne–allenones for highly diastereoselective synthesis of hexacyclic 1-naphthols†

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Received 29th July 2020,
Accepted 27th August 2020

DOI: 10.1039/d0qo00917b

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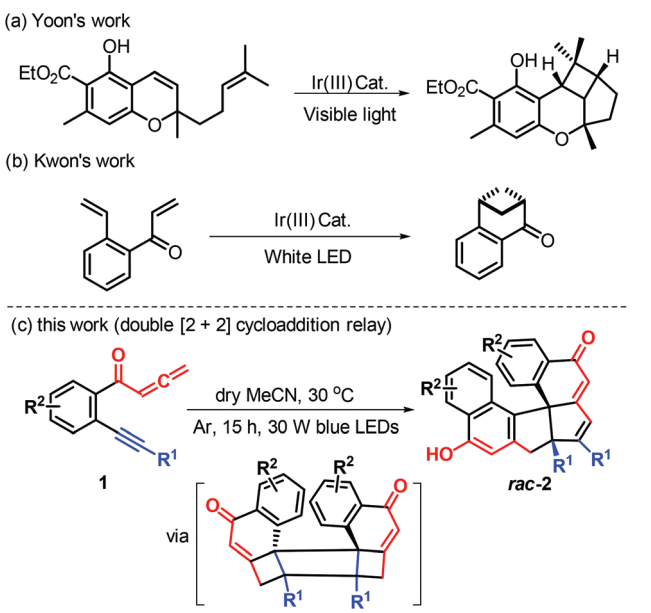
A new photoinduced photocatalyst-free energy-transfer strategy for double [2 + 2] cycloaddition relay of yne–allenones is reported for the first time and used to produce a series of hitherto unreported hexacyclic 1-naphthols with good yields and complete diastereoselectivity. The reaction pathway generates a class of complex three-dimensional structures with two all-carbon quaternary centers evolved from the planar conjugated system through cleavage and recombination of the C–C triple bond of yne–allenones, enabling the direct formation of four new rings in one step.

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Introduction

Cycloaddition reactions driven by visible light have long proven to be flourishing and powerful tools for the synthesis of various cyclic architectures under mild conditions.[1] In addition to the inherent characteristics of atom-economy, high stereoselectivity and sustainability, such reactions can directly construct intricate cyclic scaffolds that are difficult to obtain by thermal reactions. As one of the most fundamental photoinduced cycloaddition reactions, [2 + 2] cycloaddition reactions have been extensively applied to assemble numerous four-membered cyclic molecules in recent years.[2] Generally, photochemical [2 + 2] cycloadditions proceed through an energy-transfer (EnT) process, and substantial elegant studies have been published by many research groups.[3] For example, the group of Yoon reported an impressive visible light catalytic [2 + 2] cycloaddition of styrenes in the presence of an iridium (III) polypyridyl complex (Scheme 1a).[4] Kwon's group developed an energy transfer approach to obtain bridged cyclobutanes through iridium(III)-catalyzed intramodular crossed [2 + 2] cycloaddition of dienones upon visible light irradiation (Scheme 1b).[5] Despite these advances, most of these approaches developed so far require the intervention of photosensitizers including organic and metal photocatalysts, thus causing a complex work-up process. Therefore, the development of photoinduced EnT strategies for [2 + 2] cycloaddition reactions by means of the photosensitivity of substrates without additional photocatalysts is strongly encouraged and promising but full of challenges.

Recently, our group and others have developed intramolecular [2 + 2] cycloadditions and their *in situ* transformations starting from benzene-linked yne–allenones,[6] furnishing skeletally diverse naphthalen-1-ol derivatives. In particular, we recently established a new reaction mode of benzene-linked yne–allenones with alkenes or terminal alkynes, namely a



Scheme 1 Profiles of visible light induced [2 + 2] cycloaddition reactions *via* energy transfer.

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†Electronic supplementary information (ESI) available. CCDC 2015069 (**2q**). For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d0qo00917b

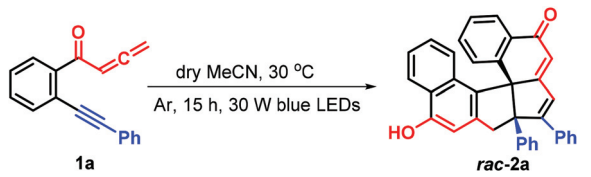
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BF_3·Et_2O-catalyzed double [2 + 2] cycloaddition relay, and the process was supported by carrying out systematic theoretical calculations.[6d] In continuation of our research in the area of double [2 + 2] cycloaddition relay, we became interested in developing such transformations by continuously using yne–allenones as starting materials under photocatalytic conditions. As we implemented this concept, we found that the conjugated yne–allenones have photoabsorbing property in the absorption band range of 400–450 nm (see Fig. 2 and 3); thus we questioned whether photoinduced double [2 + 2] cycloaddition relay of yne–allenones assisted by their own photosensitivity would work, resulting in a direct, atom-economical entry to polycyclic architectures, thereby avoiding the use of photocatalysts while greatly simplifying the work-up process. To our delight, the photocatalyst-free cyclodimerization reaction proceeded smoothly through an EnT process under blue LED irradiation, leading to the formation of the hitherto unreported hexacyclic 1-naphthols in good yields, with the complete control of stereo- and regiochemistry in which two adjacent all-carbonyl stereogenic centers were constructed (Scheme 1c). Remarkably, the current photoinduced EnT strategy enabled C–C triple-bond cleavage of yne–allenones to access unprecedented pentaleno[1,2-*a*:6*a*,6-*a*′]dinaphthalen-11-ones with good substrate scope and 100% atom utilization, offering an extensively attractive approach for assembling complex molecular scaffolds. Herein we elaborate on this attractive photoinduced double [2 + 2] cycloaddition relay.

Results and discussion

At the outset of our investigation, yne–allenone **1a** was chosen as the model substrate and the reaction of **1a** was conducted in dry CH_3CN under 30 W blue LEDs and an argon atmosphere at 30 °C for 15 h. As expected, the reaction proceeded readily, cleanly affording the hexacyclic product **2a** in 65% yield (Table 1, entry 1). Then, other light sources such as white light and green light were examined. White light could drive this transformation, but its efficiency was lower than that under blue light (entry 2). However, green light proved to be ineffective as the expected target was not observed (entry 3). Subsequently, we carefully investigated the effect of the solvent by employing different dry aprotic solvents including 1,4-dioxane, acetone, tetrahydrofuran (THF), *N*,*N*-dimethylformamide (DMF), toluene, and 1,2-dichloroethane (DCE) (entries 4–9). The results revealed that these attempts did not show any improvement on the obtainable yield. Elevating the reaction temperature to 40 °C made the reaction system chaotic, lowering the yield of **2a** to 60% (entry 10). An incomplete consumption of **1a** was detected and thus a significantly reduced yield of **2a** (30%) was provided when the reaction temperature was 25 °C (entry 11). The reaction did not work and the starting material **1a** was recovered with the removal of light irradiation (entry 12). A significant drop in the yield of **2a** was observed when the reaction was carried out under air conditions, showing that O_2 is not beneficial for this photoinduced cyclodimerization (entry 13).

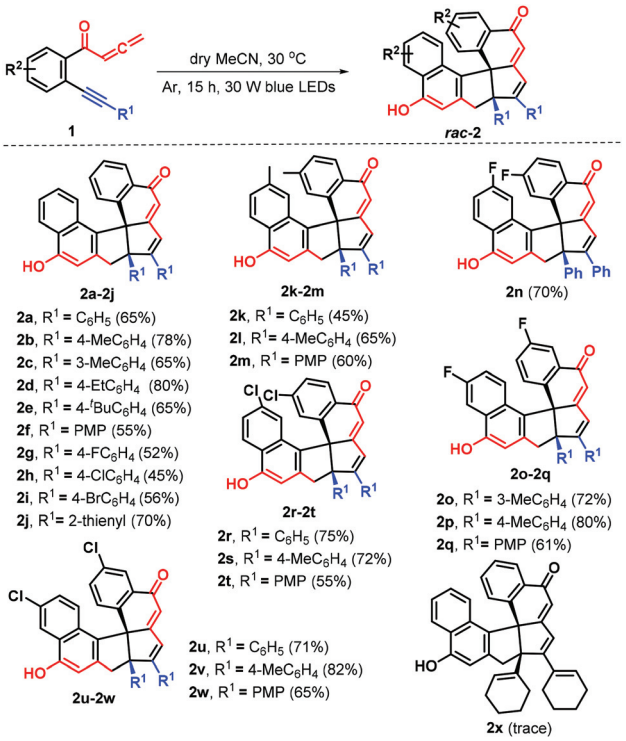
Table 1 Optimization of reaction conditions for **2a**[a]



Entry	Variation from the standard conditions	Yield[b] (%)
1	Blue LEDs	65
2	White LEDs instead of blue LEDs	45
3	Green LEDs instead of blue LEDs	Trace
4	Dry 1,4-dioxane instead of CH_3CN	50
5	Dry acetone instead of CH_3CN	55
6	Dry THF instead of CH_3CN	Trace
7	Dry DMF instead of CH_3CN	Trace
8	Dry toluene instead of CH_3CN	Trace
9	Dry DCE instead of CH_3CN	Trace
10	At 40 °C	60
11	At 25 °C	30
12	Without blue LEDs	NR
13	Under air conditions	10

[a] Reaction conditions: benzene-tethered yne–allenone (**1a**, 0.2 mmol, 1.0 equiv.), dry MeCN (2 mL), 30 °C, under argon conditions, for 15 hours. [b] Isolated yield based on substrate **1a**. N.R. = no reaction.

Having demonstrated the viability of using the yne–allenone **1a** in the photoinduced double [2 + 2] cycloaddition relay (Table 1, entry 1), we set out to systematically investigate the substrate scope of this protocol. The results are summarized in Scheme 2. Initially, the influence of substituents (R^1) in the



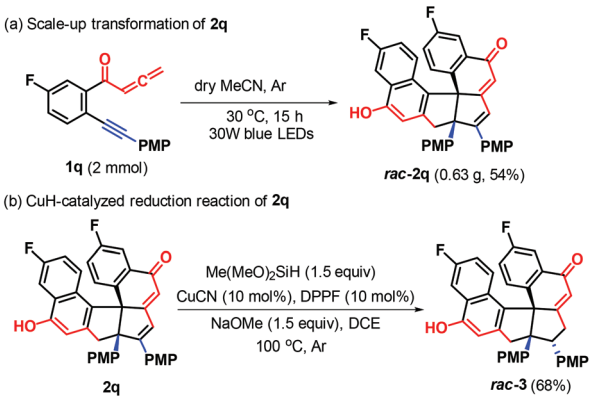
Scheme 2 Substrate scope for forming products **2**.

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arylalkynyl moiety including electronic and steric characteristics was examined. Both electron-donating (*e.g.* methyl **1b–1c**, ethyl **1d**, *tert*-butyl **1e**, methoxy **1f** (PMP = *p*-methoxyphenyl)) and electron-withdrawing (*e.g.* fluoro **1g**, chloro **1h** and bromo **1i**) groups at different positions on the phenyl ring were compatible with these photocatalytic conditions, enabling a C–C triple-bond cleavage process to furnish products **2b–2i** in 45–80% yields with complete diastereoselectivity. The presence of electron-withdrawing groups seemed to decrease the efficiency of the reaction as shown for **2g** and **2h**, in which moderate yields were obtained for both. Notably, a thienyl-substituted counterpart **1j** was found to show a high reactivity profile, thus delivering the corresponding product **2j** as a single diastereoisomer in 70% yield. Subsequently, the electronic nature of substituents (R^2) on the internal aryl ring of substrates **1** was briefly investigated. Several representative functional groups such as methyl (**1k–1m**), fluoro (**1n–1q**) and chloro (**1r–1w**) were independently installed at the C4 or C5 position of the internal arene ring and these substrates (**1k–1w**) were then subjected to the standard conditions. As anticipated, these functional groups oriented the complete diastereoselectivity to access the corresponding products **2k–2w** with comparable efficiency. However, cyclohexenyl substituted yne–allenone **1x** seems reluctant to undergo this reaction process as product **2x** was detected in a trace amount and could not be isolated. The results have been shown in Scheme 2. This protocol represents a first example of photoinduced [2 + 2] cycloaddition relay enabled C–C triple bond cleavage for the highly diastereoselective synthesis of complex three-dimensional architectures containing an unprecedented polycyclic 1-naphthol unit and opens new avenues for the development of photocatalyst-free photochemistry. The structures of these products were fully characterized by NMR spectroscopy and HRMS data (see the ESI†). The stereo-structure of **2q** was identified by single-crystal X-ray diffraction (Fig. 1).

The practical utility of this photoinduced cyclodimerization protocol was demonstrated by a scale-up experiment (Scheme 3a). When carrying out the reaction at a 2 mmol scale, the desired product **2q** was delivered smoothly in 54% yield. The experimental results showed the potential application of this strategy in actual synthesis. The presence of conjugated α,β,γ,δ-unsaturated cycloketones provides the great possibility of further modification of these hexacycles as these molecules are good 1,6-eletrophilic acceptors. For instance, treatment of **2q** with dimethoxy(methyl)silane in the presence of CuCN and 1,1′-bis(diphenylphosphino)ferrocene (DPPF) ligand as a catalytic system cleanly afforded the reductive product **3** in 68% yield under argon conditions (Scheme 3b).[7]



Scheme 3 Application of product **2q**.

To understand the mechanism of this reaction, 1.5 equivalents of 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) or butylhydroxytoluene (BHT) as a radical scavenger[8] were subjected to the reaction with **1a** under the standard conditions, and a slightly reduced yield of product **2a** was obtained (Scheme 4a), indicating that the current photocatalysis may involve an EnT process, rather than a radical pathway. Next, the conversion of **1a** into **2a** was significantly suppressed by the triplet quencher 2,5-dimethylhexa-2,4-diene (Scheme 4b),[9] which further supported the EnT process of this transformation. To further examine the possible implication of yne–allenones within the photochemical regime, we investigated the photophysical behavior of yne–allenone **1a** in dry MeCN. The inspection of the absorption spectrum indicated that the substrate **1** has photo-

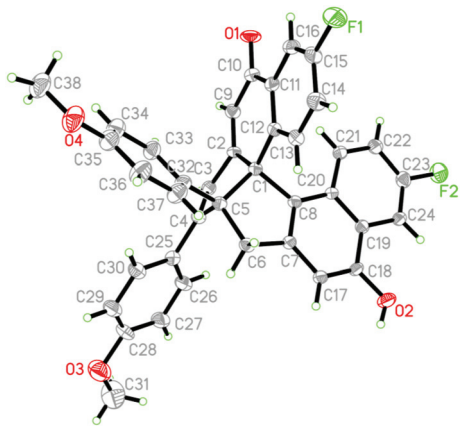
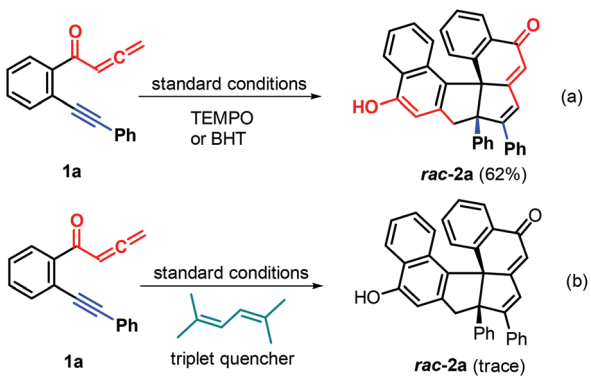


Fig. 1 X-ray structure of **2q**.



Scheme 4 Control experiments.

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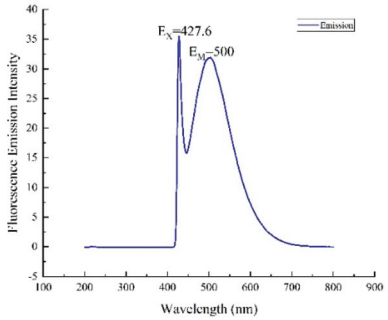


Fig. 2 Emission spectra (excitation at 427 nm) of yne–allenone **1a** (5×10^{-5} M in MeCN).

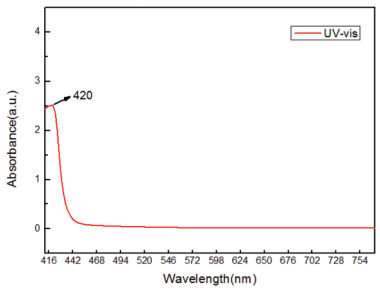
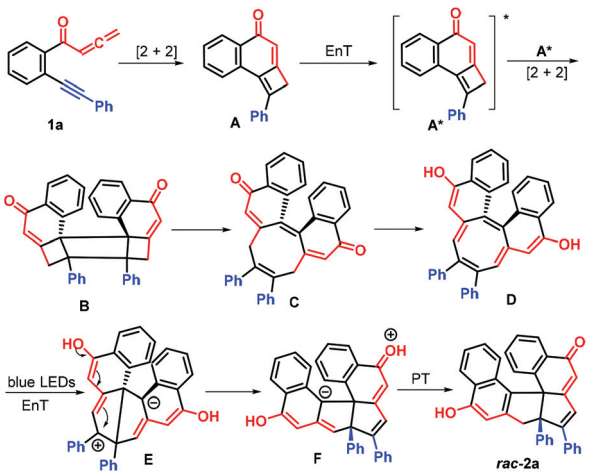


Fig. 3 Absorption spectra (excitation at 420 nm) of yne–allenone **1a** (5×10^{-5} M in MeCN).

absorbing property in the absorption band range of 400–450 nm (Fig. 2 and 3), thus the suggestive prospect arose that the direct photoexcitation of such a compound could promote intermolecular cycloadditions under blue LED irradiation.

Based on the experimental results and previous reports,[3a,10] a putative pathway leading to the formation of **2a** is proposed in Scheme 5. Initially, yne–allenone **1a** rapidly undergoes an intramolecular [2 + 2] cycloaddition to offer adduct intermediate **A**. Subsequently, under the irradiation of visible light, the ground state **A** can reach the triplet state **A*** *via* energy transfer, followed by their intermolecular [2 + 2] cycloaddition to give the intermediate **B**.[5] The ring-opening of the four-membered carbocycle, because of its ring tension, easily occurs under visible light irradiation, giving the eight-membered ring intermediate **C**, which undergoes isomerization and cycloaddition under irradiation of visible light to yield the zwitterion intermediate **E**. Subsequent isomerization and proton transfer (PT) give the final product **2a**. Although the detailed mechanism for this cyclodimerization is not clear to us yet, the current observations offer a potential and attractive pathway for the construction of complex three-dimensional architectures from staring materials with a planar conjugated system.



Scheme 5 Plausible reaction pathway.

Conclusions

In summary, starting from easily available yne–allenones, we have established a photoinduced photocatalyst-free EnT strategy for realizing double [2 + 2] cycloaddition relay, by which a wide set of hitherto unreported hexacyclic 1-naphthols with two adjacent all-carbonyl quaternary centers were efficiently synthesized in a high diastereoselective and atom-economical fashion. The reaction process is considered to proceed through a double [2 + 2] cycloaddition relay, C–C triple bond cleavage and hydrogen atom transfer sequence. The photocatalysis enjoys the complete control of regio- and diastereochemistry, good compatibility with different types of substituents and mild reaction conditions as well as 100% atom utilization and high annulation efficiency (formation of up to four new rings). Further investigations regarding the mechanism as well as synthetic application are underway in our laboratory.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

We are grateful for financial support from the NSFC (No. 21871112 and 21971090), the Graduate Education Innovation Project of Jiangsu Province (No. KYCX20_2265) and the Top-Notch Academic Programs Project of Jiangsu Higher Education Institution (TAPP).

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